

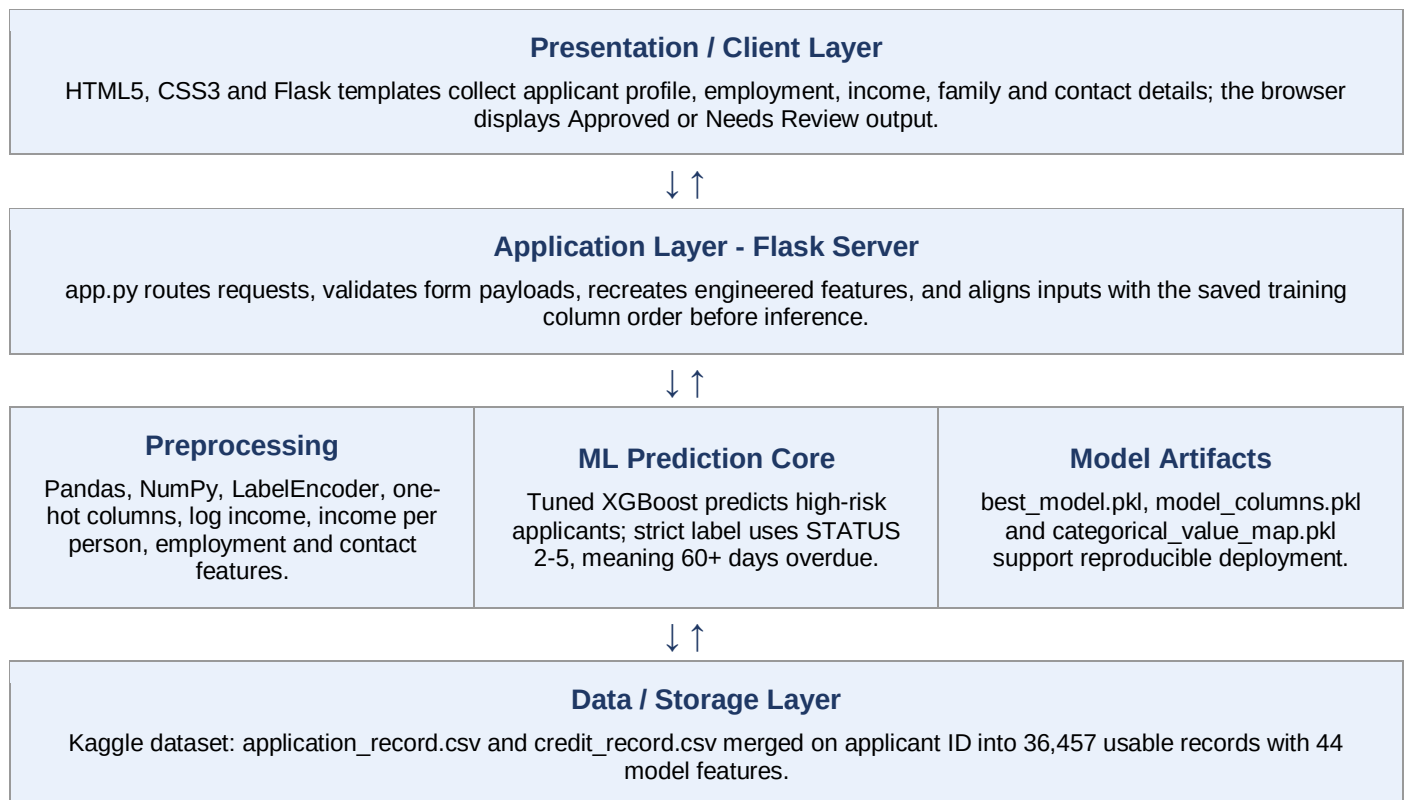
Solution Architecture

Date	15/07/2026
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Technical architectural flow for a Flask-based credit card approval prediction system using cleaned application and credit repayment history data.

Three-Tier Architectural Layout



Component Description Table

Component Name	Role in Solution	Technologies Used
Frontend + Flask	Collects applicant data, runs the request flow, and renders the approval decision.	HTML5, CSS3, Flask
Preprocessing Pipeline	Encodes categorical fields, remaps rare occupations to "Other", and derives engineered features (log income, income per person, contact score) before inference.	Pandas, NumPy, scikit-learn (LabelEncoder)

Tuned XGBoost Model	Scores default risk from engineered demographic, income, and repayment-history features.	Python, scikit-learn, XGBoost, joblib
Model Artifacts	Persists the trained model, expected column order, and categorical value mappings for reproducible, consistent inference.	joblib (best_model.pkl, model_columns.pkl, categorical_value_map.pkl)
Data / Storage Layer	Source dataset merged and cleaned into the final training table used to fit the model.	Kaggle dataset (application_record.csv, credit_record.csv), Pandas